

Harsath Kaliappa Gounder Thangavel

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Education

York University

Bachelor of Science in Computer Science (Honours)

Toronto, Canada

January 2020 – December 2024

Work Experience

Privex

Software Engineer

October 2023 – January 2024

London, United Kingdom

- Developed an AI customer service agent using RAG techniques on a self-hosted LLM to handle common customer inquiries related to technical issues, products, and services.
- Developed REST APIs in Flask for querying and training the LLM, integrated with Discord APIs to allow customers to query the AI agent and staff to train it with new data on demand.
- Utilized vector databases and MySQL for training data management. Developed scripts to remove existing data from the database and to add new data from the website's documentation.

CogniPet

Software Engineer

April 2019 – January 2020

Zürich, Switzerland

- Developed Pet Breed Detection system by training a deep learning model using the Xception neural network and transfer learning techniques in Keras to recognize 120 breeds of dogs with 93% accuracy.
- Implemented REST APIs in Flask for model inference and integrated them with CogniPet's Android and iOS apps using Amazon Web Services (AWS).
- Experimented with video-based methods for pet identification by collecting data from dogs and analyzing features from multiple frames.

Projects

Jix Interpreter | C Interpreter

- Developed an interpreter for a dynamically typed programming language with support for basic functional programming.
- Implemented fault-tolerant parsing and error handling, dynamic types, arrays, loops, functions with nested blocks and recursion, built-in functions, conditional statements, and control flow statements.
- Added functional programming support, including first-class functions, higher-order functions, and function pointers.
- Tested using automated tests written in Jix and integrated them with CI/CD pipelines.

Prism Coverage | Java Code Coverage Engine

- Developed a code coverage engine using ANTLR4 for a programming language written in Java.
- Implemented control flow coverage (statement, decision, condition, path) criteria.
- Implemented data-flow coverage (All-Defs, All-DU-Paths, All-P-Uses) criteria.
- Generated visual HTML reports with color-coded coverage details and summaries for control-flow and data-flow coverage.

Cap Containers | C Containers Library

- Developed a highly generic, header-only containers library in C.
- Implemented LRU cache, vectors with iterators, sets, maps, hash-tables, queues (circular, dynamic, fixed, priority), stack, and lists.
- Implemented thread-safe variants with cross-platform support (Windows, Linux, and macOS).
- Tested using automated tests and documented with Doxygen.

SOCKS5 Handler | C++ SOCKS5 Library

- Developed a library implementing RFC 1928 and RFC 1929 for a SOCKS5 proxy client with username/password authentication and remote DNS resolution support.

Technical Skills

Languages: Python, C++, C, Java, Go, JavaScript, SQL, Perl, Bash, PHP, HTML/CSS, $\text{\LaTeX}$
Technologies: Linux/Unix, MySQL/PostgreSQL, Django/Flask, JWTs, CI/CD pipelines, Neo4j
Developer Tools: Git, Docker, LXC/LXD, GDB/LLDB, Amazon Web Services